



## Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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### DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

2026-2027 Trimester - 4

#### Project Abstract Submission Form

|            |            |                                                           |
|------------|------------|-----------------------------------------------------------|
| Sec No- 10 | Team No- 1 | Project Title: Event and Room Occupancy Management System |
|------------|------------|-----------------------------------------------------------|

#### 1. Abstract

The Event and Room Occupancy Management System is a web-based application designed to help students and college administrators efficiently manage room availability, event scheduling, and participant registrations within a college campus. Students often face difficulty finding available rooms during college hours, while events such as hackathons, workshops, seminars, and competitions may experience unexpected increases in participants due to spot registrations. These situations can lead to room allocation conflicts, overcrowding, and inefficient utilization of available spaces.

The proposed system will provide a user-friendly web interface through which users can view available rooms, check room capacity and schedules, register for events, and access relevant event information. Administrators will be able to create and manage events, allocate rooms, monitor registrations, and track expected and actual occupancy. The system will also help identify scheduling conflicts and capacity issues, allowing administrators to make better decisions regarding room allocation.

The application will use a React-based frontend, a Node.js and Express.js backend, and a MySQL relational database for storing and managing system information. The database will maintain relationships between users, rooms, events, registrations, bookings, and schedules while ensuring data consistency and integrity. The system aims to improve campus resource utilization, reduce scheduling conflicts, provide better visibility of room availability, and offer a centralized platform for efficient event and room management.

#### 2. Team Details

| Roll Number(s) | Name of the Student | Signature |
|----------------|---------------------|-----------|
| 2520090123     | J. Geya Reddy       |           |
| 2520090001     | Kamtam Priyanka     |           |

### **Frameworks / Platforms / Software Used**

- **React.js** – Frontend framework for developing the web interface
- **Node.js** – Backend runtime environment
- **Express.js** – Backend framework for developing REST APIs
- **MySQL** – Relational database for storing users, rooms, events, registrations, bookings, and occupancy data
- **MySQL Workbench** – Database design, SQL development, and database management
- **Postman** – Testing and debugging REST APIs
- **Visual Studio Code** – Development environment
- **Git & GitHub** – Version control and source-code management

**Guide Name and Signature (remarks if any)** \_\_\_\_\_